

Window Function

Name: Nguyen Ho Ngoc Minh Student ID: 11247201 Class: DS66B

Lab 01: Film Ranking (The Basics)

Interface showing a SQL query for Lab 01: Film Ranking (The Basics) executed in a PostgreSQL environment.

Query:

```
-- Lab 01
WITH RankedFilms AS (
  SELECT c.name AS category, f.title AS film_title, f.length AS film_len,
         DENSE_RANK() OVER (PARTITION BY c.name ORDER BY f.length DESC, f.title) AS rn
  FROM film f
  JOIN film_category fc ON f.film_id = fc.film_id
  JOIN category c ON c.category_id = fc.category_id
)
SELECT * FROM RankedFilms WHERE rn <= 3
```

Data Output:

	category character varying (25)	film_title character varying (255)	film_len smallint	rn bigint
1	Action	Darn Forrester	185	1
2	Action	Worst Banger	185	2
3	Action	Casualties Encino	179	3
4	Animation	Gangs Pride	185	1
5	Animation	Pond Seattle	185	2
6	Animation	Sons Interview	184	3
7	Children	Fury Murder	178	1
8	Children	Wrong Behavior	178	2
9	Children	Empire Malkovich	177	3
10	Classics	Conspiracy Spirit	184	1
11	Classics	Beasts Control	176	2

Total rows: 48 Query complete 00:00:00.148 CRLF Ln 4, Col 79

Lab 02: Growth Analysis (Time Series)

Interface showing a SQL query for Lab 02: Growth Analysis (Time Series) executed in a PostgreSQL environment.

Query:

```
--Lab 02
WITH MonthlyRevenue AS (
  SELECT
    DATE_TRUNC('month', payment_date) AS sales_month,
    SUM(amount) AS total_revenue
  FROM payment
  GROUP BY DATE_TRUNC('month', payment_date)
),
RevenueWithLag AS (
  SELECT
    sales_month,
    total_revenue,
    LAG(total_revenue, 1) OVER (ORDER BY sales_month) AS previous_month_revenue
  FROM MonthlyRevenue
)
SELECT
  TO_CHAR(sales_month, 'YYYY-MM') AS month,
  total_revenue,
  previous_month_revenue,
  ROUND(
    (total_revenue - previous_month_revenue) * 100.0 / NULLIF(previous_month_revenue, 0),
    2
  ) AS mom_growth_rate_pct
FROM RevenueWithLag
ORDER BY sales_month;
```

Data Output Messages Notifications

14	--Lab 02
15	WITH MonthlyRevenue AS (
16	SELECT
17	DATE_TRUNC('month', payment_date) AS sales_month,
18	SUM(amount) AS total_revenue
19	FROM payment
20	GROUP BY DATE_TRUNC('month', payment_date)
21	),
22	RevenueWithLag AS (
23	SELECT
24	sales_month,
25	total_revenue,
26	LAG(total_revenue, 1) OVER (ORDER BY sales_month) AS previous_month_revenue
27	FROM MonthlyRevenue
28	)
29	SELECT
30	TO_CHAR(sales_month, 'YYYY-MM') AS month,
31	total_revenue,

month	total_revenue	previous_month_revenue	mom_growth_rate_pct
1	2007-02	8351.84	[null]
2	2007-03	23886.56	8351.84
3	2007-04	28559.46	23886.56
4	2007-05	514.18	28559.46

Total rows: 4 Query complete 00:00:00.172 CRLF Ln 38, Col 22

Lab 03: Customer Lifetime Value (Running Total)

41	--Lab 03
42	SELECT
43	customer_id,
44	payment_id,
45	payment_date,
46	amount AS transaction_amount,
47	SUM(amount) OVER (
48	PARTITION BY customer_id
49	ORDER BY payment_date, payment_id
50	ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
51	) AS cumulative_clv
52	FROM payment
53	ORDER BY customer_id, payment_date;

customer_id	payment_id	payment_date	transaction_amount	cumulative_clv
1	1	2007-02-14 23:22:38.996577	5.99	5.99
2	1	2007-02-15 16:31:19.996577	0.99	6.98
3	1	2007-02-15 19:37:12.996577	9.99	16.97
4	1	2007-02-16 13:47:23.996577	4.99	21.96
5	1	2007-02-18 07:10:14.996577	4.99	26.95
6	1	2007-02-18 12:02:25.996577	0.99	27.94
7	1	2007-02-21 04:53:11.996577	3.99	31.93
8	1	2007-03-01 07:19:30.996577	4.99	36.92
9	1	2007-03-02 14:05:18.996577	3.99	40.91

Total rows: 14596 Query complete 00:00:00.200 CRLF Ln 53, Col 36

Lab 04: The "Top-N" Filter Challenge

Query Query History Scratch Pad X

```

57 WITH CustomerCountrySpend AS (
58 -- Step 1: Aggregate total spend by Customer and Country
59 SELECT co.country_id, co.country, c.customer_id, c.first_name, c.last_name,
60        SUM(p.amount) AS total_spend
61 FROM payment p
62 JOIN customer c ON p.customer_id = c.customer_id
63 JOIN address a ON c.address_id = a.address_id
64 JOIN city ci ON a.city_id = ci.city_id
65 JOIN country co ON ci.country_id = co.country_id
66 GROUP BY co.country_id, co.country, c.customer_id, c.first_name, c.last_name
67 ),
68 RankedSpend AS (
69 -- Step 2: Apply RANK() partitioned by Country
70 SELECT country_id, country, customer_id, first_name, last_name, total_spend,
71        RANK() OVER (
72            PARTITION BY country_id
73            ORDER BY total_spend DESC
74        ) AS spend_rank
75 FROM CustomerCountrySpend
76 )
77 -- Step 3: Select top 1 customer per Country
78 SELECT
79     country,
80     customer_id,
81     first_name || ' ' || last_name AS customer_name,
82     total_spend
83 FROM RankedSpend
84 WHERE spend_rank = 1

```

Data Output Messages Notifications

Total rows: 108 Query complete 00:00:00.314 CRLF Ln 55, Col 3

Query Query History Scratch Pad X

```

57 WITH CustomerCountrySpend AS (
58 -- Step 1: Aggregate total spend by Customer and Country
59 SELECT co.country_id, co.country, c.customer_id, c.first_name, c.last_name,
60        SUM(p.amount) AS total_spend
61 FROM payment p
62 JOIN customer c ON p.customer_id = c.customer_id
63 JOIN address a ON c.address_id = a.address_id
64 JOIN city ci ON a.city_id = ci.city_id

```

Data Output Messages Notifications

Showing rows: 1 to 108 Page No: 1 of 1

	country character varying (50)	customer_id integer	customer_name text	total_spend numeric
1	Afghanistan	218	Vera Mccoy	67.82
2	Algeria	176	June Carroll	151.68
3	American Samoa	320	Anthony Schwab	47.85
4	Angola	528	Claude Herzog	93.80
5	Anguilla	381	Bobby Boudreau	99.68
6	Argentina	560	Jordan Archuleta	129.71
7	Armenia	41	Stephanie Mitch...	118.75
8	Austria	266	Nora Herrera	113.74
9	Azerbaijan	334	Raymond Mcwh...	108.75
10	Bahrain	590	Seth Hannon	108.76
11	Bangladesh	21	Michelle Clark	146.68
12	Belarus	144	Clara Shaw	189.60

Total rows: 108 Query complete 00:00:00.314 CRLF Ln 55, Col 3